Module 06: Learning — Epistemic Growth, Niche Construction, and Self-Transformation

Learning Objectives

1. Explain how learning in Active Inference is formalized as updating the parameters of the generative model (structure learning).
2. Connect the concept of niche construction to the Active Inference view that agents shape their environments to make them more predictable.
3. Evaluate the philosophical implications of learning as self-transformation — the agent that learns is no longer the same agent.

Introduction

Learning is the process by which experience changes the agent. In traditional epistemology, learning is the accumulation of justified true beliefs. In behavioral psychology, it is the modification of response patterns through reinforcement. Active Inference offers a deeper account: **learning is the updating of the generative model's parameters so that future predictions are more accurate**. This is not merely acquiring information but *changing the very structure of the mind*.

Key Concepts

1. Parameter Learning and Structure Learning

Active Inference distinguishes two forms of learning:

- **Parameter learning**: Updating the concentration parameters of existing model components (e.g., learning that a particular observation is more likely than previously expected). This corresponds to sharpening Dirichlet priors.
- **Structure learning**: Adding or removing components of the generative model itself. This is implemented through **Bayesian Model Reduction (BMR)** — analytically

comparing the evidence for different model structures and keeping the one that best balances accuracy and complexity.

Philosophically, parameter learning is like refining one's vocabulary; structure learning is like learning a new language. The latter is more transformative and more philosophically interesting.

2. Niche Construction and the Co-Evolution of Agent and Environment

Agents do not merely adapt to their environments — they also **construct** them. A beaver builds a dam; a human builds a city. In Active Inference, niche construction is the environmental complement of learning: the agent acts on its environment to make it more predictable, thereby reducing future surprise.

This connects to the philosophical concept of **Umwelt** (von Uexküll) — the species-specific world that each organism inhabits. The Umwelt is not a pre-given environment but a co-construction of organism and world. Laland et al. (2015) argue that niche construction is a neglected dimension of evolution, alongside natural selection.

3. Learning as Self-Transformation

If the agent *is* its generative model (as Active Inference suggests), then changing the model is changing the agent. Learning is not the addition of content to a fixed container but the **transformation of the container itself**. This resonates with Bildung in the German philosophical tradition — education as self-formation, not mere information transfer.

It also raises a paradox: the agent that needs to learn is not yet the agent who has learned. Pre-learning, the agent cannot fully grasp what it will become. This connects to Kierkegaard's concept of "the leap" and to Thomas Kuhn's account of paradigm shifts in science.

Applications

- **Education**: If learning is self-transformation, education is not the transfer of facts from teacher to student but the guided restructuring of the student's generative model.

- **Psychotherapy**: Therapeutic change, in Active Inference terms, involves restructuring maladaptive generative models — not merely providing insight but changing the patterns of prediction and precision that sustain distress.

Conclusion

Learning in Active Inference is not a passive accumulation of data but an active, transformative process that changes the agent's very identity. This has profound implications for education, therapy, and the philosophy of personal identity. Module 07 examines how agents share their models with others — the problem of communication.